Xiaoyu Ou

PhD Candidate in Electrical and Electronic Engineering
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Available for a postdoctoral position from June 2027.

Research Profile

Theoretical research: Finite-blocklength (FBL) communications, with a focus on transceiver design and resource allocation for multiuser MIMO and ultra-reliable low-latency communications (URLLC); Next-generation multiple access, including RSMA and NOMA.
Practical implementation: USRP X310 testbed development using Python/UHD and MATLAB, including device synchronization, channel estimation, multiuser precoding, and over-the-air testing.

Education

University of Bristol, Bristol, UK Sep. 2023 – Present
PhD in Electrical and Electronic Engineering
Fully funded by the University of Bristol
Thesis: *Transceiver Design and Resource Allocation for Finite-Blocklength Multiuser MIMO Communications*.
Supervisors: Dr. Shuping Dang and Prof. Angela Doufexi.
Expected thesis submission: November 2026; expected viva: before February 2027.

Chongqing University of Posts and Telecommunications, Chongqing, China Sep. 2020 – Jul. 2023
MSc in Information and Communication Engineering
Supervisor: Prof. Xianzhong Xie.

Chongqing University of Posts and Telecommunications, Chongqing, China Sep. 2016 – Jul. 2020
BEng in Electronic Science and Technology

Research Experience

- **Multiuser transmission and interference management.** Studied uplink RSMA with finite SIC, massive MIMO-NOMA with group-level SIC, and dual-polarized transmission without SIC. Developed power-control and receiver-design algorithms for these systems. *[J1, J3, J5, C1]*
- **Resource allocation for short-packet communications.** Developed algorithms for user scheduling, power allocation, and rate control to improve effective throughput in multicarrier RSMA. Compared metadata and payload encoding schemes in massive MIMO, with power, receive combining, and blocklength optimization to reduce the worst-user block error rate. *[J2, J4]*
- **Resource-allocation mismatch between infinite and finite blocklength.** Studied the sum-rate loss when allocations designed for infinite blocklength are used for FBL transmission.*[S1]*

Teaching Experience

Teaching Assistant, University of Bristol Feb. – May 2026
Helped supervise the undergraduate group project *Covert Communications via Motion States of Unmanned Aerial Vehicle*.

Honors and Awards

Outstanding Master's Thesis of Chongqing 2024
University of Bristol PhD Scholarship in Electrical and Electronic Engineering 2023
Outstanding Graduate of Chongqing 2023
National Scholarship, Ministry of Education of China 2022

Academic Service and Technical Skills

Journal reviewer: IEEE Transactions on Communications; IEEE Transactions on Wireless Communications; IEEE Transactions on Vehicular Technology; IEEE Communications Letters; IEEE Wireless Communications Letters.
Tools and methods: MATLAB; Python; convex optimization; theoretical analysis and modeling.

Future Research Interests

Theoretical research: Finite-blocklength information theory and its potential applications; fast optimization algorithms for wireless signal processing.
Practical implementation: FPGA implementation of signal processing and optimization algorithms, and USRP experiments to test their performance and compare the results with theoretical predictions.

Publications

Journal Articles

[J1] **X. Ou**, S. Dang, and A. Doufexi, "Uplink RSMA with Dual-Polarized Multi-User MIMO over Correlated Rician Channels in Finite Blocklength Regime," *IEEE Transactions on Communications*, early access, 2026.

[J2] **X. Ou**, S. Dang, and A. Doufexi, "Reliable Uplink Transmission for Low-Throughput Massive MIMO Networks With Finite Blocklength," *IEEE Transactions on Communications*, vol. 74, pp. 2716–2730, 2026.

[J3] **X. Ou**, S. Dang, Z. Ren, and A. Doufexi, "Uplink Power Control for Massive MIMO-NOMA With Group-Level SIC in Massive URLLC Services," *IEEE Transactions on Wireless Communications*, vol. 24, no. 10, pp. 8505–8519, Oct. 2025.

[J4] **X. Ou**, S. Dang, Z. Ren, and A. Doufexi, "Joint User Scheduling, Power Allocation, and Rate Control for MC-RSMA in URLLC Services," *IEEE Transactions on Vehicular Technology*, vol. 74, no. 6, pp. 9210–9223, Jun. 2025.

[J5] **X. Ou**, S. Dang, and A. Doufexi, "Sum Spectral Efficiency Maximization for Uplink Multi-Layer RSMA With Finite SIC," *IEEE Wireless Communications Letters*, vol. 13, no. 12, pp. 3409–3413, Dec. 2024.

[J6] **X. Ou**, X. Xie, H. Lu, and H. Yang, "Channel Blocklength Minimization in MU-MISO Nonorthogonal Multiple Access for URLLC Services," *IEEE Systems Journal*, vol. 18, no. 1, pp. 36–39, Mar. 2024.

[J7] **X. Ou**, X. Xie, H. Lu, and H. Yang, "Resource Allocation in MU-MISO Rate-Splitting Multiple Access With SIC Errors for URLLC Services," *IEEE Transactions on Communications*, vol. 71, no. 1, pp. 229–243, Jan. 2023.

[J8] **X. Ou**, X. Xie, H. Lu, H. Yang, and H. Tang, "Energy-Efficient Resource Allocation for Short Packet Transmission in MISO Multicarrier NOMA," *IEEE Transactions on Vehicular Technology*, vol. 71, no. 12, pp. 12797–12810, Dec. 2022.

[J9] W. B. Abbas, **X. Ou**, G. Hilton, S. Dang, A. Doufexi, and M. A. Beach, "Antenna Selection and Array Directivity Tradeoff in Massive MIMO Systems With Uniform Rectangular Arrays," *IEEE Transactions on Communications*, vol. 74, pp. 7285–7299, 2026.

[J10] W. Wang, **X. Ou**, Z. Ren, W. B. Abbas, S. Dang, A. Doufexi, and M. A. Beach, "Analysis on Energy Efficiency of RIS-Assisted Multiuser Downlink Near-Field Communications," *IEEE Transactions on Communications*, vol. 73, no. 11, pp. 12441–12455, Nov. 2025.

[J11] Y. Chen, H. Yang, **X. Ou**, Y. Jiang, and Z. Xiong, "Anti-Jamming Resource Allocation for Integrated Sensing and Communications Based on Game-Guided Reinforcement Learning," *IEEE Wireless Communications Letters*, vol. 14, no. 1, pp. 223–227, Jan. 2025.

[J12] X. Xie, **X. Ou**, H. Lu, and Q. Huang, "Joint Uplink and Downlink Resource Allocation in NOMA for End-to-End URLLC Services," *IEEE Communications Letters*, vol. 25, no. 12, pp. 3942–3946, Dec. 2021.

Conference Paper

[C1] **X. Ou**, S. Dang, and A. Doufexi, "Dual-Polarized MIMO for Uplink Rate-Splitting Transmission with Finite Blocklength," in *Proc. IEEE 36th International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, Istanbul, Türkiye, Sep. 2025, pp. 1–6. (Invited paper.)

Manuscript Under Review

[S1] **X. Ou**, S. Dang, and A. Doufexi, "On Mismatch of Sum-Rate Maximization Between Infinite and Finite Blocklength Regimes," under review at *IEEE Transactions on Signal Processing*, 2026.